SOVEREIGN VOICE AGENT FOR LOAN COLLECTIONS

1. The Problem

Indian NBFCs and banks collectively place an estimated **50M+ outbound collection calls every month**, at a typical cost of **₹60–80 per human-agent call** once telephony, agent salary, and supervision overhead are included. Three structural problems compound the cost:

- Capacity, not willingness, is the constraint. Human collections teams cannot call every overdue account within the first few days of delinquency the window where a reminder is most effective and least adversarial so a meaningful share of early-stage overdue accounts simply go uncontacted.
- Vernacular borrowers are underserved. Tier 2/3 accounts are disproportionately Hindi, Marathi, and regional-language speakers who code-mix with English mid-sentence. English-first or single-language IVR systems degrade badly on exactly these calls.
- Human-agent variance is a compliance risk. Collections language is regulated (RBI Fair Practices Code); a stressed or undertrained agent ad-libbing wording about debt or legal follow-up is a real, recurring source of complaints and regulatory exposure.
- Agent attrition in collections BPOs frequently exceeds 30–40% annually, so retraining cost and inconsistent call quality are recurring, not one-time, problems.

2. Why AI

A voice agent addresses the capacity constraint directly; it can attempt every overdue account within hours of due-date breach, 24/7, without headcount scaling linearly with call volume. Three properties matter more than raw automation:

- **Consistency at scale.** Every borrower hears the same compliance-approved wording: no ad-libbing, no agent-to-agent variance.
- **Language-native, not language-translated.** The end user is often a first-generation smartphone owner who is far more comfortable speaking Hindi or Marathi naturally including code-mixed with English than navigating an app or a DTMF menu tree.
- **Faster, cleaner escalation.** The agent's job is not to replace judgment, it is to triage. Anything that needs a human (hardship, disputes, refusals) is flagged and routed within the same call, with full context already captured, rather than losing that context in a second, disconnected call.

3. Why Sarvam

Generic frontier models (Gemini, GPT, Claude) are excellent generalists but are not built to own this specific problem. Sarvam's advantage is concentrated in exactly the dimensions that matter for a regulated Indian lender:

Dimension	Sarvam	Generic Western LLM API

Data residency	Trained, hosted, governed entirely in India	Foreign-hosted by default a compliance blocker for RBI-regulated data
Code-mixing	Native handling of Hinglish / Marlish mid-sentence switching	Multilingual support exists, but code-mixing is an edge case, not a design target
Latency architecture	Streaming STT + streaming TTS built for telephony-grade real-time voice	Voice is typically a wrapper around a text-first model
Access continuity	Not subject to foreign export-control decisions	Foreign access to advanced models has already been restricted once in 2026

In plain terms: ***"can a bank legally and reliably deploy this"*** is answered by Sarvam's architecture before the conversation about model quality even starts.

4. Architecture Summary

In plain terms, one call works like this see the accompanying architecture diagram for the technical version:

- The borrower speaks. Sarvam converts speech to text in real time, automatically detecting whether they are speaking Hindi, Marathi, or a mix with English.
- A language model classifies what the borrower needs: a payment promise, a hardship request, a dispute; it does not decide what to say.
- What the borrower hears is always a pre-approved, compliance-reviewed script, spoken back in their own language. The model's classification decides which script line plays; it never writes new wording live.
- If a human is needed, the system automatically creates a ticket in the operations queue with the reason, the extracted detail, and priority already filled in before the call even ends.

This POC demonstrates the full loop end-to-end: **conversation → classification → scripted response → CRM ticket → ops dashboard.**

5. ROI / Business Case

Illustrative model for a mid-size NBFC — assumptions stated explicitly; substitute real portfolio figures for an actual proposal.

Assumption	Human agent	AI voice agent
Tier-1 reminder calls per month (assumed portfolio)	100,000	100,000
Cost per call (all-in — see breakdown below)	₹80	₹2.60–3.10
Share automatable without human involvement (Tier-1, low-complexity)	—	~70%
Blended monthly cost (70% AI-handled, 30% human escalation at ₹80)	**₹8,000,000**	**₹2,600,000**

Estimated monthly savings: **~₹5.4M (~68%)** → **~₹65M annualized**, before counting revenue-side effects.

Cost-per-call breakdown, all figures sourced from sarvam.ai/api-pricing.

- **Sarvam API cost — ₹1.57/call:** STT ₹30/hr × ~45s speech (₹0.38) + TTS ₹3/1,000 chars × ~385 chars (₹1.16) + Sarvam-105B-Conversations, 3 classification calls (₹0.04). 3-turn happy path; busier calls cost modestly more.

- **Telephony — ₹0.70–1.00/call (assumption):** typical domestic SIP/PSTN rate, ~2-min call. Not a Sarvam cost the production transport layer.

- **Hosting/compute — ₹0.30–0.50/call (assumption):** amortized server cost at pilot scale.

Two effects beyond direct cost, each individually significant:

- **Capacity, not just cost.** Accounts that were never called due to human-agent capacity limits can now be reached on day one of delinquency, which is the single highest-leverage moment in the recovery curve.

- **Compliance risk reduction.** Scripted, auditable calls reduce exposure to RBI Fair Practices Code violations that stem from agent improvisation.

6. Limitations & Next Steps

What this POC does not yet do

- Runs over a browser microphone, not a real telephone line, production requires a SIP trunk (Twilio/Exotel).
- Covers 2 languages (Hindi, Marathi) end-to-end; the platform supports 22+, and rollout would be prioritized by portfolio language mix.
- The conversation flow covers the primary collections journey (promise-to-pay, hardship, dispute) not every edge case a live portfolio will surface.

- The CRM integration is a mock ticket store production that would write directly into the lender's actual CRM/LOS.

Proposed 90-day rollout

Phase	Scope
Days 0–30	Telephony integration (SIP); native-speaker QA + legal sign-off on scripts; pilot on 1 branch (~5,000 accounts).
Days 30–60	Real CRM/LOS integration replacing the mock ticket store; add 2–3 more portfolio languages; post-call analytics via Sarvam's batch STT + diarization pipeline.
Days 60–90	Full regional rollout; expand flow (settlement bands, legal-notice); ops SLA dashboard; A/B recovery-rate vs. human baseline.

Note - Working demo, code, and architecture diagram accompany this document. See README.md for setup instructions and the demo video link.